

Solving systems of non linear equations

$$\underline{f}(\underline{x}) = \underline{0} \Leftrightarrow \begin{cases} f_1(x_1, x_2, \dots, x_n) = 0 \\ f_2(x_1, x_2, \dots, x_n) = 0 \\ \vdots \\ f_n(x_1, x_2, \dots, x_n) = 0 \end{cases}$$

Given $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$

find $\alpha \in \mathbb{R}^n$ s.t. $f(\alpha) = \underline{0} \in \mathbb{R}^n$

Newton method for nonlinear systems

for scalar equations the Newton Method reads

given $x^{(0)}$

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$$

$$\begin{cases} f'(x^{(k)}) \delta x^{(k)} = -f(x^{(k)}) \\ x^{(k+1)} = x^{(k)} + \delta x^{(k)} \end{cases}$$

Generalizing to $\mathbb{R}^n$ we can define the method

$$\begin{cases} J(x^{(k)}) \delta x^{(k)} = -f(x^{(k)}) \\ x^{(k+1)} = x^{(k)} + \delta x^{(k)} \end{cases}$$

$J(x^{(k)})$ is the so called Jacobian matrix of f evaluated at $x^{(k)}$

$$J(x^{(k)}) = [J_{ie}(x^{(k)})]$$

$$J_{ie}(x^{(k)}) = \frac{\partial f_i}{\partial x_e}(x^{(k)})$$

For example in $\mathbb{R}^2$

$$f(x) = \begin{bmatrix} x_1^2 + \sin(x_2) \\ x_2^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} x_1^2 + \sin(x_2) = 0 \\ x_2^2 = 0 \end{cases}$$

f_1 (pointing to the first component) and f_2 (pointing to the second component)

$$J(x) = \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \frac{\partial f_1(x)}{\partial x_2} \\ \frac{\partial f_2(x)}{\partial x_1} & \frac{\partial f_2(x)}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 & \cos(x_2) \\ 0 & 2x_2 \end{bmatrix}$$

Newton method for systems

- At each step

1) Evaluate $J^{(k)} = J(x^{(k)})$

2) Solve the linear system

$$J^{(k)} \delta x^{(k)} = -f(x^{(k)})$$

3) Update $x^{(k)}$ as

$$x^{(k+1)} = x^{(k)} + \delta x^{(k)}$$

The vector-Newton method is

- local

$$\exists \delta > 0 \text{ s.t. if } \|\alpha - x^{(k)}\| < \delta$$

$$\lim_{k \rightarrow \infty} \|\alpha - x^{(k)}\| = 0 \quad \text{local convergence}$$

- Order 2

If the method converges and

$J(x)$ is invertible for $x = \alpha$ and smooth

then

$$\frac{\|\alpha - x^{(k+1)}\|}{\|\alpha - x^{(k)}\|^2} \leq C \quad \text{for } k > \bar{k}$$

Stopping criteria

• residual

$$\|f^{(k)}\| < \varepsilon$$

• increment

$$\|x^{(k+1)} - x^{(k)}\| < \varepsilon$$

Computational cost

$$\text{CPU TIME} = \# \text{ iter} \times (C_{\text{cos}} + C_{\text{se}})$$

compute J
?
(> solve $Jx = b$)

To reduce the cost one can use so called quasi-Newton methods that use approximations of J that are

- easier to construct $\Rightarrow$ small C_{cos}
- easier to invert $\Rightarrow$ C_{se}

For example : update J only every p iteration

Broyden method

$$B_k \delta x^{(k)} = -f(x^{(k)})$$

$$x^{(k+1)} = x^{(k)} + \delta x^{(k)}$$

$$B_{k+1} = B_k + \frac{f(x^{(k+1)}) (\delta x^{(k)})^T}{(\delta x^{(k)})^T \delta x^{(k)}}$$